

HR EMPLOYEE ATTRITION & RETENTION ANALYSIS

Data Analyst Internship Project Report

1. INTRODUCTION

Employee attrition is an important Human Resources challenge because high employee turnover can increase recruitment costs, reduce productivity, and affect workforce stability. This project focuses on analyzing employee attrition patterns and identifying the major factors associated with employee turnover.

The project uses HR employee data to understand attrition across different departments, job roles, overtime status, age groups, tenure groups, job satisfaction levels, and income. Python was used for data analysis and machine learning, while Power BI was used to create an interactive dashboard for presenting important HR insights.

The main objective of the project is to understand the causes and patterns of employee attrition and develop a machine learning model that can help identify employees who may be at higher risk of leaving.

2. ABSTRACT

This project presents an end-to-end analysis of employee attrition using the IBM HR Analytics Employee Attrition dataset. The dataset contains **1,470 employee records and 35 columns**.

The data was loaded and analyzed using Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Data quality checks identified **0 missing values and 0 duplicate records**.

Exploratory Data Analysis was performed to study attrition based on department, overtime, job role, age, tenure, job satisfaction, and other employee characteristics. Feature engineering was performed by creating an AttritionFlag, AgeGroup, and TenureGroup.

A **Logistic Regression** classification model was developed to predict employee attrition. The model achieved an accuracy of **74.83%**, with a precision of **34.83%**, recall of **65.96%**, and F1 score of **45.59%**.

The analysis found an overall employee attrition rate of **16.12%**. The highest attrition was observed in the **Sales department (20.63%)**, among **Sales Representatives (39.76%)**, employees below **25 years (35.77%)**, and employees with **0–1 years of tenure (34.88%)**.

The results were also visualized using Power BI to provide an interactive view of employee attrition and support HR decision-making.

3. TOOLS USED

Tool / Technology	Purpose
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Python	Data analysis and machine learning
Pandas	Data loading, cleaning and data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Exploratory data visualization
Scikit-learn	Machine learning and model evaluation
Logistic Regression	Employee attrition classification
Power BI	Interactive HR analytics dashboard
Excel	Supporting data analysis and reporting

4. STEPS INVOLVED IN BUILDING THE PROJECT

Step 1: Data Loading and Cleaning

The HR employee dataset was imported into Python using Pandas. The dataset contains **1,470 rows and 35 columns**.

Data quality checks were performed to identify missing values and duplicate records.

Results:

- Missing Values: **0**
- Duplicate Rows: **0**

Therefore, the dataset was suitable for further analysis after the basic data quality checks.

Step 2: Feature Engineering

The Attrition variable was converted into a binary AttritionFlag, where employees who left the organization were represented by 1 and employees who stayed were represented by 0.

Two additional features were created:

- **AgeGroup** – groups employees according to age.

- **TenureGroup** – groups employees according to years at the company.

These features were used to perform segment-level attrition analysis.

Step 3: Exploratory Data Analysis

Exploratory Data Analysis was performed to understand employee attrition patterns.

The analysis included:

- Attrition by Department
- Attrition by Overtime
- Attrition by Job Role
- Attrition by Age Group
- Attrition by Tenure
- Attrition by Job Satisfaction
- Employee income analysis

The overall employee attrition rate was **16.12%**, with **237 employees** recorded as having left the organization.

The average monthly income was **6502.93**, while the average employee tenure was **7.01 years**.

Step 4: Key Attrition Findings

The analysis identified several employee groups with comparatively high attrition.

Department:

The **Sales department** recorded the highest department-level attrition rate at **20.63%**.

Job Role:

Sales Representatives had the highest job-role attrition rate at **39.76%**.

Age:

Employees below **25 years** had the highest age-group attrition rate at **35.77%**.

Tenure:

Employees with **0–1 years** at the company had the highest tenure-group attrition rate at **34.88%**.

These findings indicate that younger employees, newer employees, and certain job roles may require additional retention attention.

Step 5: Machine Learning

A **Logistic Regression classification model** was developed to predict employee attrition.

The data was divided into training and testing datasets. Numerical variables were standardized, while categorical variables were converted using one-hot encoding.

The model was trained using the employee characteristics available in the dataset.

Step 6: Model Evaluation

The Logistic Regression model produced the following results:

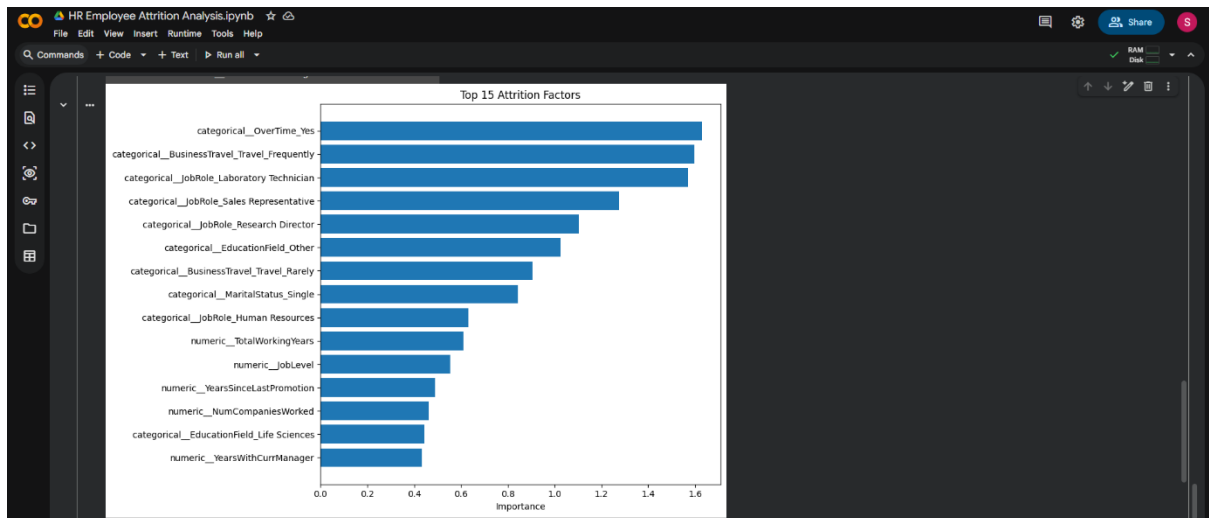
Evaluation Metric Result

Accuracy	74.83%
Precision	34.83%
Recall	65.96%
F1 Score	45.59%

A confusion matrix was also generated to evaluate the model's correct and incorrect classifications.

Feature coefficients were analyzed to identify the variables that had stronger influence on the model's attrition classification.





Step 7: Power BI Dashboard

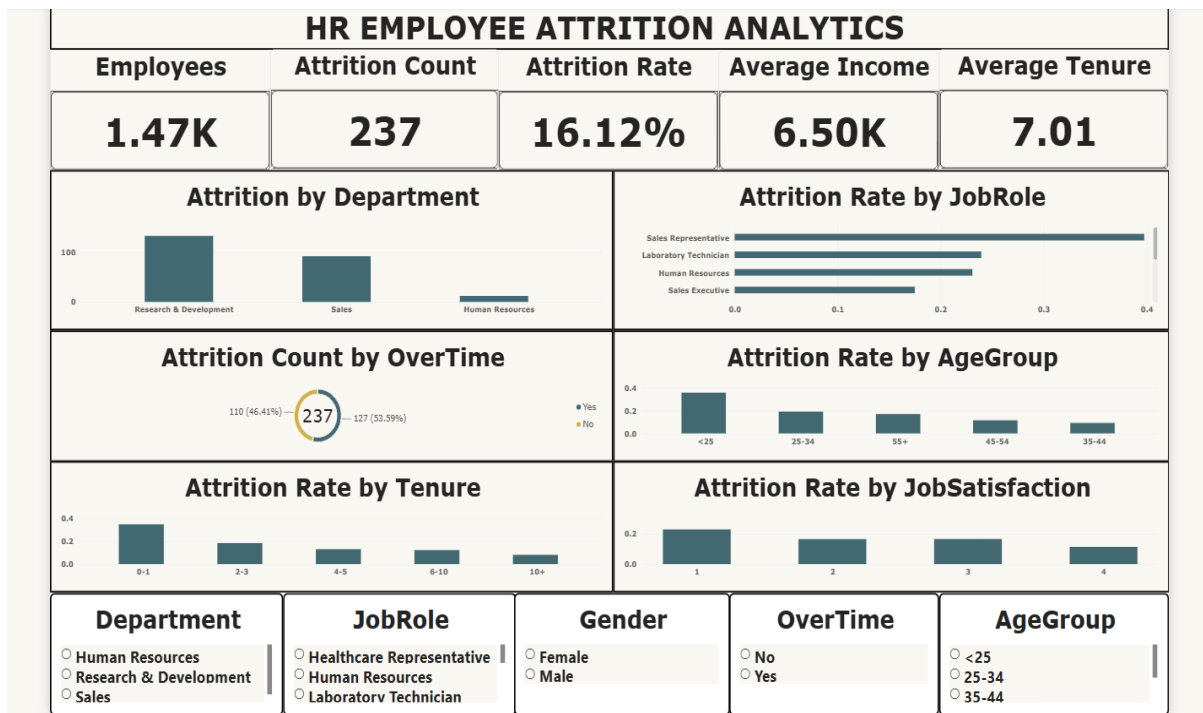
An interactive Power BI dashboard was developed to present the HR analysis visually.

The dashboard includes important KPIs such as:

- Total Employees
- Attrition Count
- Attrition Rate
- Average Income
- Average Tenure

It also provides visual analysis of attrition by department, job role, overtime, age, and tenure.

Interactive filters/slicers allow users to explore employee attrition across different employee categories.



5. BUSINESS RECOMMENDATIONS

Based on the analysis, the following recommendations can help organizations improve employee retention:

- Monitor overtime:** Employees working overtime should be monitored to identify workload and work-life balance issues.
- Improve career development:** Organizations should provide training, career progression, and internal mobility opportunities.
- Review compensation:** Compensation and salary progression should be reviewed for employee groups showing higher attrition.
- Improve employee engagement:** Regular employee feedback and engagement initiatives can help identify workplace concerns.
- Focus on high-risk groups:** Retention programs should particularly focus on high-attrition job roles, departments, younger employees, and newer employees.
- Use predictive analytics:** Machine learning can be used as a supporting tool to identify employees who may require proactive retention attention.

6. CONCLUSION

The HR Employee Attrition & Retention Analysis project demonstrates an end-to-end data analytics workflow, including data cleaning, feature engineering, exploratory data analysis, machine learning, model evaluation, feature analysis, and Power BI visualization.

The analysis identified an overall attrition rate of **16.12%** among 1,470 employees. Sales had the highest department-level attrition, Sales Representatives had the highest job-role attrition, employees below 25 years had the highest age-group attrition, and employees with 0–1 years of tenure had the highest tenure-group attrition.

The Logistic Regression model achieved **74.83% accuracy** and a **65.96% recall**, providing a useful predictive approach for identifying potential attrition cases. Combined with the Power BI dashboard, the project provides HR teams with data-driven insights that can support employee retention and workforce planning decisions.